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Q7-1 $G(s) = \frac{250}{s(0.1s+1)}$ $\quad G(s)G_c(s) = \frac{250}{s(0.1s+1)} \cdot \frac{0.05s+1}{0.0047s+1}$

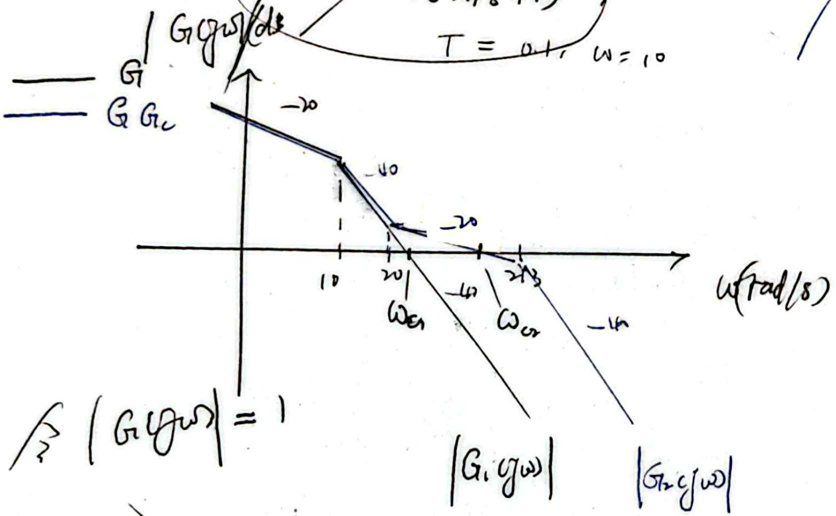
$T = 0.1, \ \omega = 10$

$T = 0.1 \quad \omega = 10$

$T = 0.0047 \quad \omega = 213$

$T = 0.05 \quad \omega = 20$

$20 \quad \frac{1}{20} \quad \frac{50}{1000} \quad 0.005 \quad 0.0047$



令 $|G(j\omega)| = 1$

$\omega_{c1} = 50\ rad/s, \quad \omega_{c2} = 125\ rad/s$

$\gamma_1 = 180 + \angle G_1(j\omega) \quad \gamma_2 = 180^\circ + \angle G_2(j\omega)$

$= 180 + (-90^\circ - \arctan\frac{0.1\omega_{c1}}{1}) \quad = 180^\circ + (-90^\circ - \arctan(\frac{0.1\omega_{c2}}{1}) + \arctan(\frac{0.05\omega_{c2}}{1}) - \arctan(\frac{0.0047\omega_{c2}}{1})$

$= 11.3^\circ \quad = 55.1^\circ$

系统② 为 系统① 加上 近似比例微分校正环节

比例微分 $\left\{ \begin{array}{l} \omega_c \nearrow \\ \gamma \nearrow \end{array} \right.$ 稳定性 $\nearrow$

Q7-2

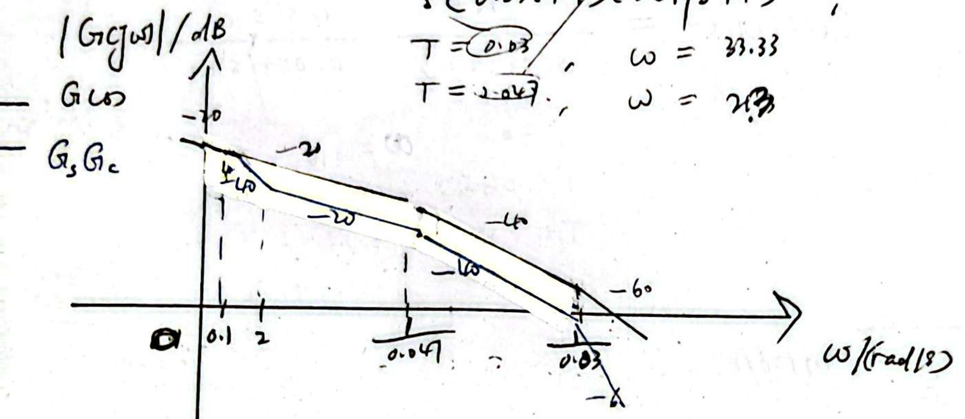
Q7-2 $G(s) = \dfrac{300}{s(0.03s+1)(0.047s+1)}$, $G_sG_c(s) = \dfrac{300(0.5s+1)}{s(10s+1)(0.03s+1)(0.047s+1)}$

$\frac{3}{100}$ $\frac{47}{1000}$

$T = 0.03$, $\omega = 33.33$

$T = 0.047$, $\omega = 21.3$

T		33.33
T		21.3
T	0.5	2
T	10	0.1



$\begin{cases} \omega_{c1} = 59 \text{ rad/s} \\ \omega_{c2} = 15 \text{ rad/s} \end{cases}$

$\begin{cases} \gamma_1 = 180 + \angle G(j\omega_{c1}) = 180^\circ - 90^\circ - \arctan(0.03\omega_{c1}) - \arctan(0.047\omega_{c1}) \\ \gamma_2 = 180^\circ + \angle G(j\omega_{c2}) = 180^\circ - 90^\circ - \arctan([\;]\omega_{c2}) - \arctan([\;]\omega_{c2}) + \arctan([\;]\omega_{c2}) - a\text{—}([\;]\omega_{c2}) \end{cases}$

$\therefore \begin{cases} \gamma_1 = -49^\circ \\ \gamma_2 = 23^\circ \end{cases}$

② 在 ① 加比例积分校正

$\begin{cases} \omega_c \searrow \\ \gamma \nearrow \end{cases}$ 降低快速性，提高稳定性

$\begin{cases} \omega_c \text{（快速性）} \\ \gamma \text{（稳定性）} \end{cases}$

Q7-20 $G_0(s) = \frac{100}{s(10s+1)}$, PID, 闭环极点 $\overset{主导}{-2\pm j1}$, -5

$$G_c(s) = \frac{K_d s^2 + K_p s + K_I}{s}$$

$G' = G G_c$

$$\frac{X_o(s)}{X_i(s)} = \frac{\frac{K_d s^2 + K_p s + K_I}{s}\cdot\frac{100}{s(10s+1)}}{1 + \qquad}$$

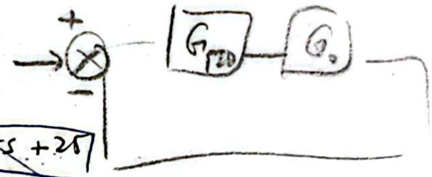
s^3

s^2

s^1

s^0

$$= \frac{10(K_d s^2 + K_p s + K_I)}{s^3 + \frac{(1+100K_d)}{10}s^2 + 10K_p s + 10K_I}$$



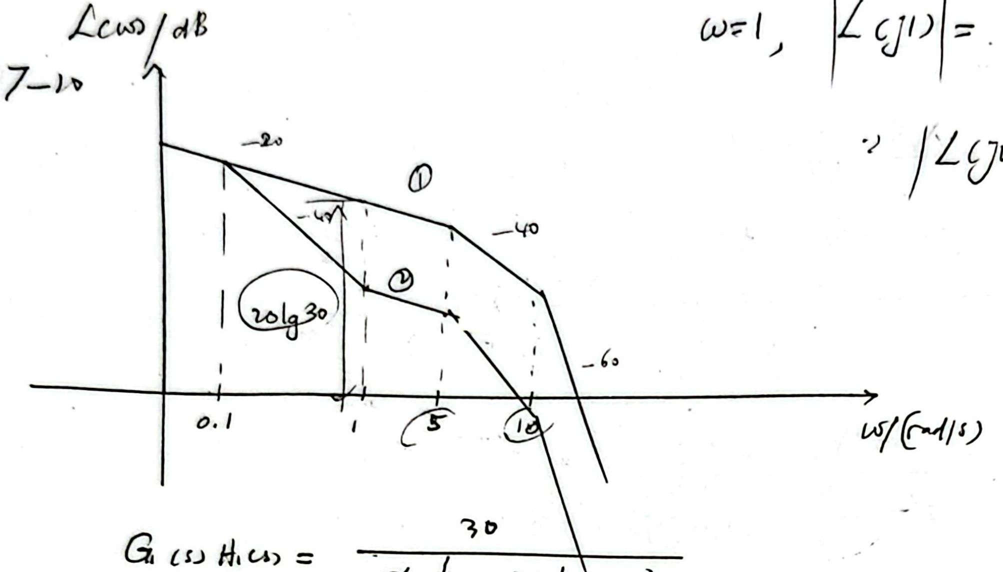
$$(s+5)(s+2-j1)(s+2+j1) = \boxed{s^3 + 9s^2 + 25s + 25}$$

~~特征方程~~ 闭环极点 $= s^3 + \frac{(1+100K_d)}{10}s^2 + 10K_p s + 10K_I$

$K_d = 0.89$, $K_p = 2.5$, $K_I = 2.5$

$$\therefore G_c = \frac{0.89s^2 + 2.5s + 25}{s}$$

Q7-10



$\omega = 1$, $|L(j1)| = 20\lg 30$

$\because |L(j\omega)| = 20\lg K$, $\therefore K = 30$

$$G_1(s)H_1(s) = \frac{30}{s(\frac{1}{5}s+1)(\frac{1}{10}s+1)}$$

$\omega_{c1} = 11$ rad/s

$\gamma_1 = 180 - \arctan(\frac{1}{5}\omega_{c1}) - \arctan(\frac{1}{10}\omega_{c1}) - 90°$

$\approx -23.3°$

$$G_2(s)H_2(s) = \frac{30(s+1)}{s(\frac{1}{5}s+1)(\frac{1}{10}s+1)(\frac{1}{0.1}s+1)}$$

$\omega_{c2} = 3$ rad/s

$\gamma_2 = 180° - 90° + \arctan(\frac{\omega_2}{T}) - \arctan$

$= 25.8°$

$\therefore$

$$G_J(s) = \frac{G_2H_2}{G_1H_1} = \frac{s+1}{10s+1}$$

$\therefore G_1H_1 \xrightarrow{G_J} G_2H_2$

Q5 2阶系统是否可以用 PI//PD 校正，任意配置极点？

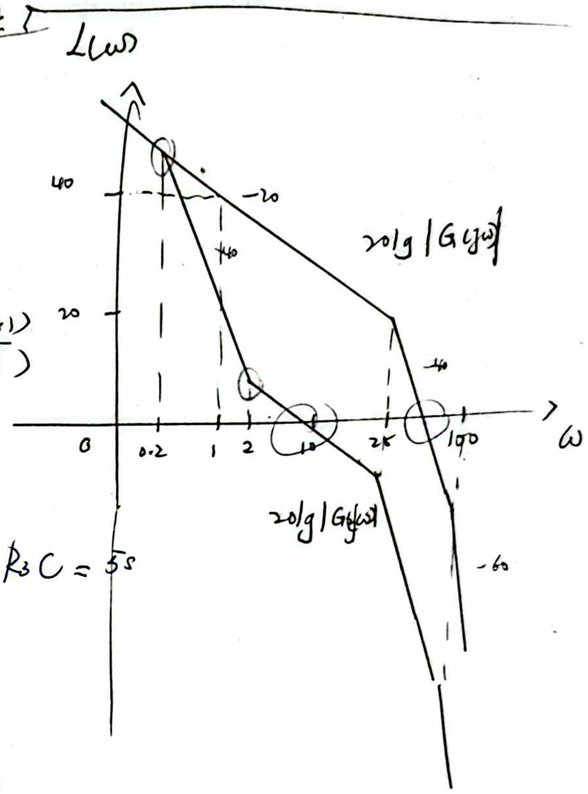
2阶系统 PID 校正后是一个闭环三阶系统，

有3个极点，故 K_P, K_I, K_D ~~中只有两个能自由调节时~~，都自由才能任意配置极点。

若 K_P, K_I, K_D 中仅两个能自由调节，无法任意配置3个极点。

Q6 为什么一般不单纯用微分校正？

① 纯微分校正 放大高频噪声



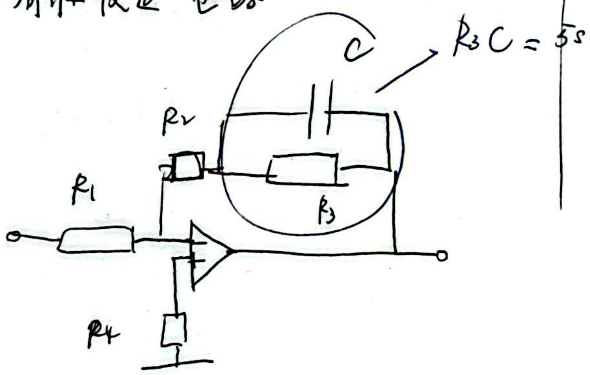
∵ $K_1=40$,

Q7 $G_1(s)=\dfrac{40}{s(\frac{1}{25}s+1)(\frac{1}{100}s+1)}$

$G_2(s)=\dfrac{40}{s(\frac{1}{25}s+1)(\frac{1}{100}s+1)}\cdot\dfrac{(0.5s+1)}{(5s+1)}$

∴ $G_c(s)=\dfrac{0.5s+1}{5s+1}$　　$\dfrac{T_2s+1}{T_1s+1}$

② 设计有源校正电路



$R_3C=5s$,

$\dfrac{R_2R_3}{R_2+R_3}C=0.5s$

$\dfrac{R_2+R_3}{R_1}=1$

$G_c(0)=\dfrac{0+1}{0+1}=1$ /（静 —）

③ 此滞后校正不影响静态增益

即不影响的稳态误差(ess)，

通过快速性换取了稳定性

$\omega_{c2}<\omega_{c1}$（截止频率↓）

(2)

$$G_p(s) = \frac{20}{(0.1s+1)(0.001s^2+0.01s+1)}$$

极点：-10，$-5 \pm 5\sqrt{39}j$

应用PID消去 $-5 \pm 5\sqrt{39}j$，使之成I阶系统

如：$\frac{5}{s(0.1s+1)}$

消去震荡项：$G_c(s) = \frac{K_D s^2 + K_p s + K_I}{s} = \frac{A(0.001s^2+0.01s+1)}{s}$

$\cancel{A=\frac{1}{4}}$，$G_1 \cdot G_c = G_2$

$20A = 5$，$A = \frac{1}{4}$

∴ $K_D = \frac{1}{4000}$，$K_p = \frac{1}{400}$，$K_I = \frac{1}{4}$